COCOBOD

Ghana Cocoa Board

Telephone Exchange Management System

System Documentation, Audit Report &
Comprehensive Enhancement Implementation Plan

Technology Stack: PHP | React | MySQL

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1. System Overview

1.1 Introduction

The COCOBOD Telephone Exchange Management System is a web-based platform built to digitise, centralise, and streamline the management of the Ghana Cocoa Board's internal telephone exchange operations. The system replaces manual telephone directory and maintenance request processes with a structured, role-based digital workflow accessible from any browser.

The system is currently operational and running on a local development environment. It is built on a robust, industry-standard technology stack comprising PHP for server-side logic, React for a dynamic and responsive frontend interface, and MySQL as the relational database engine.

1.2 Technology Stack

Layer	Technology	Purpose
Frontend	React.js	Dynamic UI, component rendering, state management
Backend	PHP	API endpoints, business logic, authentication, session management
Database	MySQL	Data persistence, relational queries, reporting data source
Architecture	MVC / REST API	Separation of concerns, RESTful communication between React and PHP

1.3 User Roles

The system currently implements three distinct user roles, each with a dedicated dashboard and scoped access privileges:

Role	Access Level	Primary Responsibility
Administrator	Full System Access	System configuration, user management, reporting, oversight of all operations
Receptionist	Operational Access	Managing telephone directory, logging incoming calls, handling extension requests
Lab Technician / Maintenance	Technical Access	Receiving and managing maintenance requests, logging faults, updating repair status

2. Current System Audit

2.1 Audit Prompt — Understanding the Current System

The following is the comprehensive audit prompt to be used with an AI-assisted development tool (such as Integravity AI) to perform a full read-through and understanding of the current codebase before any enhancements are applied:

AUDIT INSTRUCTION:

You are performing a comprehensive audit of the COCOBOD Telephone Exchange Management System. Read the entire codebase thoroughly. Map every file, every database table, every API route, and every dashboard component. Understand the role-based access control, the data flow between React and PHP, and the MySQL schema. DO NOT make any changes. Report only.

2.2 Current System Architecture

2.2.1 Database Schema (Current)

The system maintains a MySQL database with the following core tables:

Table Name	Key Fields	Purpose
users	id, name, email, role, password_hash	Authentication and role management for all three user types
extensions	id, extension_no, department, staff_name, status	Telephone extension directory and assignment records
call_logs	id, caller, extension, timestamp, duration, notes	Log of all calls handled through the exchange
maintenance_requests	id, fault_type, location, priority, status, assigned_to	Fault reports and maintenance job tracking
departments	id, name, head, extension_count	COCOBOD department registry linked to extensions

2.2.2 API Communication Flow

React components communicate with the PHP backend via RESTful API calls. The standard communication pattern is:

- React component mounts → triggers an API fetch to PHP endpoint
- PHP validates session and user role → queries MySQL → returns JSON response
- React updates component state → re-renders the UI with received data
- Form submissions follow POST → PHP validates and inserts/updates → returns success/error JSON

💡 *All API routes are PHP files returning JSON. Session-based authentication guards each endpoint against unauthorised access.*

2.3 Dashboard Functionality Audit

2.3.1 Administrator Dashboard

The Administrator dashboard provides full system oversight with the following current capabilities:

- User Management — Create, edit, deactivate, and assign roles to system users
- Extension Directory Management — Add, update, and remove telephone extensions
- Department Management — Manage COCOBOD department records
- Maintenance Request Oversight — View all submitted requests and their current status
- Call Log Review — Access the full historical call log across all receptionists
- System Reports — Basic reporting on call volumes and maintenance activity
- System Settings — Configuration of system parameters

2.3.2 Receptionist Dashboard

The Receptionist dashboard is the operational front-line interface with the following current capabilities:

- Extension Lookup — Search the telephone directory by name, department, or extension number
- Call Logging — Record incoming and outgoing calls with caller details, extension, duration, and notes
- Maintenance Request Submission — Log a new fault or maintenance request on behalf of a caller
- Directory Browse — Browse all active extensions by department
- My Call History — View the receptionist's own logged call history

2.3.3 Lab Technician / Maintenance Dashboard

The Maintenance Technician dashboard supports the technical team with the following current capabilities:

- Maintenance Request Queue — View all open maintenance requests assigned to them or unassigned
- Request Detail View — Open a specific request to view full fault details, location, and priority
- Status Update — Update the status of a request: Pending → In Progress → Resolved
- Job Notes — Add technical notes to a maintenance job
- Completed Jobs History — View a log of previously resolved requests

3. Comprehensive Enhancement Implementation Plan

The following enhancements are designed to transform the current operational system into a production-ready, enterprise-grade platform suitable for formal presentation to COCOBOD leadership. Enhancements are grouped by category and prioritised by impact.

3.1 Security Enhancements

3.1.1 Authentication & Session Security

Enhancement	Description	Priority
Password Hashing Upgrade	Implement bcrypt (cost factor 12) for all password storage. Add salting per user.	**CRITICAL**
Session Hardening	Regenerate session ID on login. Set secure, httpOnly, and SameSite cookie flags. Implement session timeout after inactivity.	**CRITICAL**
CSRF Protection	Add CSRF tokens to all forms and API mutations. Validate on every POST request.	**HIGH**
Rate Limiting	Limit login attempts to 5 per 15 minutes per IP. Show lockout message and auto-unlock after 30 minutes.	**HIGH**
Two-Factor Authentication	Optional 2FA via email OTP for Administrator role. Admin can enforce it system-wide.	**MEDIUM**
Audit Log	Log every login, logout, data change, and deletion with user ID, timestamp, IP address, and action detail.	**HIGH**

3.1.2 Data Security

- Input Validation & Sanitisation — All user inputs validated server-side in PHP using prepared statements. No raw SQL queries permitted.
- SQL Injection Prevention — Enforce PDO with parameterised queries across all database interactions.
- XSS Protection — Escape all output with htmlspecialchars(). Implement Content Security Policy headers.
- HTTPS Enforcement — Configure for HTTPS in production. Redirect all HTTP to HTTPS.
- Role-Based Access Control (RBAC) enforcement — Every PHP endpoint checks the session role before processing.

3.2 Administrator Dashboard Enhancements

3.2.1 Enhanced User Management

- User Profiles — Full profile view per user with activity summary, last login, and account status
- Bulk User Operations — Bulk activate, deactivate, or reassign roles
- User Activity Monitor — Real-time view of who is currently logged in

- Password Reset Tool — Admin-triggered password reset link sent to user email
- Role Permission Matrix — Visual table showing what each role can and cannot do

3.2.2 Advanced Analytics Dashboard

Widget / Chart	What It Shows
KPI Summary Cards	Total extensions, active calls today, open maintenance requests, resolved this week — at a glance
Call Volume Line Chart	Daily/weekly/monthly call volume trends with date range filter
Department Call Distribution	Pie/donut chart showing call distribution across COCOBOD departments
Maintenance Request Heatmap	Visual heatmap of fault frequency by location/department over time
Request Resolution Time	Average time from request submission to resolution, by technician and by fault type
Extension Status Board	Live board showing Active / Faulty / Unassigned extensions across all departments

3.2.3 Reports & Export

- PDF Report Generation — One-click export of call logs, maintenance reports, and directory listings as formatted PDFs
- Excel/CSV Export — Export any data table to CSV or Excel for offline analysis
- Scheduled Reports — Configure automated weekly/monthly email reports to designated recipients
- Audit Trail Report — Full exportable log of all system actions for compliance

3.3 Receptionist Dashboard Enhancements

3.3.1 Intelligent Extension Search

- Smart Search — Real-time search with instant results as the receptionist types, matching on name, department, extension number, and job title
- Recent Searches — Quick access panel showing the last 10 searched extensions
- Favourites — Pin frequently used extensions for one-click access
- Extension Status Indicator — Live badge on each extension card: Active (green) / Faulty (red) / Unassigned (grey)
- Click-to-Copy Extension — Single click copies the extension number to clipboard

3.3.2 Advanced Call Logging

- Quick Log Modal — Floating modal that pre-fills date/time automatically; receptionist fills caller name, extension, and notes only
- Call Duration Timer — Built-in stopwatch the receptionist can start when a call begins and stop when it ends
- Call Tagging — Tag calls by type: Enquiry / Transfer / Maintenance Request / Emergency / Other
- Bulk Log Review — Filter and search personal call log by date range, department, or tag
- Missed Call Tracking — Flag calls that could not be connected and note the reason

3.3.3 Maintenance Request Improvements

- Inline Request Form — Maintenance request form accessible directly from the extension card — pre-fills extension and department
- Priority Badge — Visual priority selector: Low / Medium / High / Emergency with colour-coded badges
- Request Tracker — Receptionist can see the current status of requests they submitted
- Notification on Resolution — Receptionist receives an in-app notification when a request they submitted is resolved

3.4 Maintenance Technician Dashboard Enhancements

3.4.1 Job Queue & Prioritisation

- Priority-Sorted Queue — Jobs automatically sorted by priority: Emergency → High → Medium → Low
- Colour-Coded Status Badges — Pending (orange), In Progress (blue), Resolved (green), Escalated (red)
- Assigned vs Unassigned Filter — Toggle between personal job queue and all unassigned jobs
- Self-Assignment — Technician can claim an unassigned job from the queue
- Job Escalation — Escalate a job to the Administrator if it requires external resource or approval

3.4.2 Job Detail & Documentation

- Rich Job Notes — Formatted notes with timestamp per update, building a chronological job history
- Photo Attachment — Upload photos of the fault before and after repair (stored in the database as file references)
- Parts Used Log — Record materials and parts used during the repair for stock tracking
- Resolution Summary — Structured resolution note including root cause, action taken, and recommendations
- Estimated Completion Time — Technician can set an estimated resolution time visible to the receptionist and admin

3.4.3 Performance Dashboard

- Personal Stats Panel — Jobs completed this week/month, average resolution time, open vs closed ratio
- Job History Calendar — Calendar view of completed and scheduled jobs

3.5 UI/UX Enhancement Plan

3.5.1 Design System

Design Element	Specification
Colour Palette	Primary: COCOBOD Dark Green (#1B5E20) | Accent: Gold (#F9A825) | Background: Off-white (#FAFAFA) | Surface: White (#FFFFFF)
Typography	Headings: Inter Bold | Body: Inter Regular | Monospace (IDs/codes): JetBrains Mono | Base size: 14px
Component Library	Tailwind CSS utility classes + custom component set: Button, Badge, Card, Modal, Table, Toast, Sidebar
Responsive Design	Mobile-first layout. Sidebar collapses to hamburger on tablet. All tables scroll horizontally on small screens.
Dark Mode	Full light/dark theme toggle stored in localStorage. All components must support both themes.
Loading States	Skeleton loaders on all data tables. Spinner on all action buttons during API calls. Disable buttons during submission.
Toast Notifications	Success (green), Error (red), Warning (amber), Info (blue) toasts — top-right, auto-dismiss after 4 seconds.

3.5.2 Advanced Table Component

All data tables across the system must be upgraded to a consistent advanced table component with the following features:

- Column Sorting — Click any column header to sort ascending/descending
- Search/Filter Bar — Inline search filters the visible table rows in real time
- Pagination — Configurable rows per page (10 / 25 / 50 / 100) with page navigation
- Column Visibility Toggle — User can show/hide specific columns
- Row Selection — Checkbox per row for bulk actions
- Export Button — Export current view (respecting filters) to CSV or PDF
- Status Badges — Colour-coded inline badges for status columns
- Action Column — Consistent View / Edit / Delete action buttons per row

3.5.3 Navigation & Layout

- Persistent Sidebar — Fixed sidebar with grouped navigation sections, active state highlighting, and collapsible sub-menus
- Breadcrumb Navigation — Breadcrumb trail on every page showing current location in the system
- Notification Bell — Top-bar bell icon with unread count badge. Dropdown shows latest notifications with timestamps.
- Global Search — Top-bar search that queries extensions, call logs, and maintenance requests simultaneously
- Quick Actions Bar — Floating action button with shortcuts for the most common actions per role
- Dashboard Cards — Metric summary cards at the top of each dashboard with icons, values, and trend indicators

4. Implementation Roadmap

4.1 Phased Implementation Plan

Phase	Focus Area	Key Deliverables	Duration
Phase 1	Security Hardening	Password hashing, session security, CSRF tokens, SQL injection prevention, audit log table	3–4 days
Phase 2	Database Upgrade	Add audit_log, notifications, file_attachments tables. Add missing columns. Write migration scripts.	2–3 days
Phase 3	UI/UX Overhaul	Design system implementation, advanced table component, sidebar, notifications bell, dark mode, responsive layout	5–7 days
Phase 4	Admin Enhancements	Analytics dashboard, KPI cards, charts, advanced reports, PDF/CSV export, user management upgrades	4–5 days
Phase 5	Receptionist Enhancements	Smart search, quick log modal, call timer, favourites, maintenance request tracker, notification on resolution	3–4 days
Phase 6	Technician Enhancements	Priority queue, job escalation, photo attachments, parts log, resolution summary, performance stats panel	3–4 days
Phase 7	Testing & QA	End-to-end testing of all roles, security penetration test basics, cross-browser testing, mobile responsiveness	2–3 days
Phase 8	Presentation Prep	Final UI polish, demo data seeding, PowerPoint preparation, live demo rehearsal	1–2 days

💡 Always run the audit prompt first (Section 2.1) before starting any phase. Complete and test each phase before moving to the next.

4.2 AI-Assisted Implementation Prompts

Use the following structured prompts with your AI development tool (Integravity AI) for each phase. Always paste phases in the SAME conversation thread sequentially — never start a new chat between phases.

Phase 1 — Security Prompt

You have completed a full audit of the COCOBOD Telephone Exchange System. Now apply Phase 1 — Security Hardening. Upgrade password storage to bcrypt (cost 12). Harden all sessions: regenerate ID on login, set httpOnly and SameSite=Strict cookie flags, implement 30-minute inactivity timeout. Add CSRF token generation and validation to all forms and POST endpoints. Replace all raw SQL with PDO parameterised queries. Add an audit_log table and log every login, logout, data insert, update, and delete with user ID, role, IP address, action, and timestamp. DO NOT break any existing functionality. Show BEFORE and AFTER for every file modified.

Phase 3 — UI/UX Prompt

Apply Phase 3 — UI/UX Overhaul. Implement the design system: primary colour #1B5E20 (COCOBOD Green), accent #F9A825, Inter font, Tailwind CSS. Build a persistent collapsible sidebar. Add a top navigation bar with notification bell (with unread count badge), global search, and user avatar. Build the advanced table component: column sorting, inline search, pagination (10/25/50/100 rows), CSV export, colour-coded status badges, and action column. Implement dark mode toggle stored in localStorage. Add skeleton loaders on all data tables. Add toast notifications (success/error/warning/info). Apply consistently across all three dashboards. Preserve all existing functionality.

5. Enhanced Database Schema

The following additional tables are required to support the enhanced functionality. These are additive — they do not alter existing tables unless explicitly noted.

5.1 New Tables Required

Table	Key Fields	Purpose
audit_log	id, user_id, role, action, table_affected, record_id, ip_address, created_at	Complete system activity audit trail
notifications	id, recipient_id, type, message, link, is_read, created_at	In-app notification bell events
file_attachments	id, entity_type, entity_id, file_path, file_name, uploaded_by, created_at	Photo and document attachments for maintenance jobs
user_preferences	id, user_id, theme, rows_per_page, favourite_extensions	Per-user UI preferences (dark mode, pagination settings)
scheduled_reports	id, report_type, frequency, recipient_email, last_sent, is_active	Automated report scheduling configuration
maintenance_parts	id, request_id, part_name, quantity, cost, logged_by, created_at	Parts and materials used in each repair job

5.2 Columns to Add to Existing Tables

Table	New Column(s)	Purpose
users	last_login_at, login_attempts, locked_until, two_fa_enabled	Support login tracking, rate limiting, and 2FA
maintenance_requests	estimated_resolution_at, escalated_to, resolved_at, resolution_note	Track full job lifecycle including escalation and resolution documentation
call_logs	call_type, is_missed, resolved	Support call tagging and missed call tracking
extensions	last_verified_at, notes	Track when an extension was last verified as working

6. System Presentation Guide

This section is your complete reference for preparing the PowerPoint presentation and live demonstration for COCOBOD stakeholders.

6.1 Recommended Presentation Structure

#	Slide Title	Key Content	Duration
1	**Title Slide**	System name, COCOBOD branding, presenter name, date	30 sec
2	**Problem Statement**	What challenges does COCOBOD's telephone exchange currently face without this system?	1 min
3	**Solution Overview**	What the system does, who uses it, and the three-role structure	1 min
4	**Technology Stack**	PHP + React + MySQL diagram. Why this stack is robust, scalable, and maintainable.	1 min
5	**System Architecture**	High-level architecture diagram: Browser → React → PHP API → MySQL	1 min
6	**Administrator Demo**	Live walk-through: login as admin, show analytics dashboard, user management, reports	3 min
7	**Receptionist Demo**	Live walk-through: search an extension, log a call, submit a maintenance request	2 min
8	**Technician Demo**	Live walk-through: view job queue, update a job status, add resolution notes	2 min
9	**Security Features**	Highlight: role-based access, session security, audit trail, CSRF protection	1 min
10	**Key Benefits**	Efficiency gains, error reduction, accountability through audit trail, scalability	1 min
11	**Roadmap**	Future enhancements: mobile app, SMS integration, PABX system API integration	1 min
12	**Q&A**	Open floor for questions	5 min

6.2 Key Talking Points for COCOBOD Audience

Why This System Matters

- Eliminates the manual telephone directory — no more paper lists that go out of date
- Creates accountability — every call and maintenance request is logged with the handler's name and timestamp
- Reduces maintenance response time — technicians receive instant notifications and can update status in real time
- Provides management visibility — the admin analytics dashboard shows exactly how the exchange is performing
- Fully digital audit trail — every action in the system is logged for compliance and review

Technical Strength Points

- PHP is COCOBOD's preferred server technology — no new infrastructure required
- React provides a modern, fast UI without full page reloads — users get an app-like experience
- MySQL is enterprise-grade and already used across COCOBOD's infrastructure
- Role-based access control ensures staff only see and do what they are authorised for
- The system is designed to scale — adding new departments, users, or even new roles requires no structural changes

6.3 Demo Preparation Checklist

- Seed the database with realistic demo data: at least 20 extensions across 8 departments, 50 call log entries, 15 maintenance requests in various statuses
- Create one demo account per role: admin@cocobod.gh / receptionist@cocobod.gh / technician@cocobod.gh — all with memorable passwords for the demo
- Test the full demo flow at least twice before the presentation
- Have a backup: screenshots of every key screen in case the live system cannot be accessed
- Test the system on the presentation laptop/projector — confirm the UI renders correctly at the display resolution
- Prepare answers for likely questions: How is data backed up? Can it work offline? Can it integrate with the current PABX?

7. Quick Reference — System Summary

A condensed overview of the complete system for quick reference during preparation and presentation.

7.1 What the System Does — In One Paragraph

The COCOBOD Telephone Exchange Management System is a secure, web-based platform that digitises the management of COCOBOD's internal telephone exchange. It enables receptionists to search and manage the extension directory and log all calls digitally; maintenance technicians to receive, track, and resolve telephone fault requests with full job documentation; and administrators to oversee all operations through a real-time analytics dashboard, manage users and departments, and generate compliance reports. Built on PHP, React, and MySQL, the system provides role-based access control, a complete audit trail, and a modern responsive interface designed for daily operational use.

7.2 Role Capabilities — At a Glance

Capability	Receptionist	Technician
Search telephone directory	☑ Full access	☑ Read only
Log calls	☑ Own logs	✗ No access
Submit maintenance requests	☑ Can submit	✗ Receives only
Update job status	✗ No access	☑ Full access
View analytics dashboard	✗ No access	✗ Personal stats only
Manage users	✗ No access	✗ No access
Export reports	✗ No access	✗ No access

NOTE: The Administrator has full access to all capabilities listed above, plus user management, system configuration, report export, and the complete audit trail.

7.3 System Data Flow Summary

- Receptionist logs a call → stored in call_logs table → visible to Admin in real time
- Receptionist submits maintenance request → stored in maintenance_requests → technician notified instantly
- Technician updates job status → maintenance_requests table updated → receptionist notified of resolution
- Any login, data change, or deletion → audit_log entry created automatically
- Admin views analytics → queries aggregate data from call_logs + maintenance_requests + extensions

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